

C1

Input: standard input
Output: standard output

Time Limit: 1 second
Memory Limit: 32 MB

Triangle Trouble

Siyam is learning about triangles in geometry class. He learns that a valid triangle must satisfy the triangle inequality rule, where the sum of the lengths of any two sides must be greater than the third. Excited, he decides to check if three given side lengths can form a valid triangle or not.

Input

The input contains multiple lines, each with three integers A, B, and C — the lengths of the triangle sides. Input ends with a line `0 0 0` which should not be processed.

Output

For each test case, print:

- `Valid triangle` if the sides can form a triangle
- `Invalid triangle` otherwise

Constraints

- $1 \leq A, B, C \leq 1000$

Sample Input

```
3 4 5
1 2 3
5 5 9
0 0 0
```

Sample Output

```
Valid triangle
Invalid triangle
Valid triangle
```

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C2

Input: standard input
Output: standard output

Time Limit: 1 second
Memory Limit: 32 MB

Prime Digit Sum

Shuvo is interested in numbers whose **sum of digits is a prime number**. Can you help him determine this?

Input

First line contains integer T — the number of test cases.
Each of the next T lines contains a single non-negative integer N .

Output

For each test case, print:

- `Yes` if the sum of digits of N is a prime number.
- `No` otherwise.

Constraints

- $0 \leq N \leq 10000$

Sample Input

```
4
23
17
100
29
```

Sample Output

```
Yes
Yes
No
No
```

Explanation

- $2 + 3 = 5 \rightarrow$ prime
- $1 + 7 = 8 \rightarrow$ not prime
- $1 + 0 + 0 = 1 \rightarrow$ not prime
- $2 + 9 = 11 \rightarrow$ prime

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C3

Character Balance

Input: standard input
Output: standard output

Time Limit: 1 second
Memory Limit: 32 MB

Given a string of lowercase letters, we define it as **balanced** if the number of vowels is equal to the number of consonants.

Input

The input contains multiple test cases, one string per line.
The input ends with the word `STOP`, which should not be processed.

Output

For each test case:

- Print `Balanced` if number of vowels equals number of consonants.
- Otherwise, print `Not Balanced`.

Constraints

- String contains only lowercase letters and length ≤ 100

Sample Input

```
apple  
kite  
aeiou  
bcdfg  
STOP
```

Sample Output

```
Not Balanced  
Balanced  
Not Balanced  
Not Balanced
```